Byung Hyun Lee

AI Researcher

Multimodal LLMs & Selective Knowledge Updates

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KEY HIGHLIGHTS

Aligning **multimodal LLMs across modalities**, post-training them efficiently, and **updating them without forgetting**.

CROSS-MODAL ALIGNMENT

Lift weak modalities with a strong modality

- **DCAT** (NeurIPS 2026 Spotlight): vision → audio / video MLLMs in closed form
- **+29.55%** audio & **+24.92%** video in average gain

OMNIMODAL LLM AT SCALE

Post-training an any-to-any omni model

- HyperCLOVA X 8B Omni / 32B Think at NAVER Cloud
- **Stable at ~2× faster inference** · **+20%** training throughput

POST-TRAINING · RL & SELF-DISTILLATION

Cheaper, more robust post-training

- **S^3-OPSD**: AIME24 **+10.0**, up to **13× faster** than GRPO
- **MOSAR** (VLA): robust to instruction deviations

SELECTIVE KNOWLEDGE UPDATES

Expand or erase knowledge without forgetting

- **Continual learning**: ICCV 2023, AAAI 2024 Oral, ICCV 2025
- **Concept erasure**: ICLR 2025, CVPR 2025, CVPR 2026 (**2,072** concepts)

KEY HIGHLIGHTS

8 **Top-tier AI papers**
NeurIPS · CVPR · ICCV · ICLR · AAAI

7 **First / co-first author**
among top-tier papers

4 **ICLR 2027 submissions**
3 as co-first author

7 **Industry & gov. projects**
3 as project leader

Research Recognition

- NeurIPS 2026 **Spotlight** · AAAI 2024 **Oral** · CVPR 2026 **Compute Transparency Champion Award**
- MICCAI 2025 REG **2nd place (team leader)** · CVPR 2026 **Outstanding Reviewer** · Qualcomm Fellowship **Finalist**

Projects

- **Project leader** — Gigapixel histopathology whole slide image, world VLA model
- **Samsung Research** (robot grasping), **US Air Force · IITP** (human-machine teaming)

Industry Experience

- **NAVER Cloud**, National AI Foundation Model Project — contributed to **HyperCLOVA X 8B Omni / 32B Think**

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04 Expanding: Continual Learning Adding knowledge without forgetting

05 Erasing: Concept Erasure Unlearning targets, keeping the rest

06 Domain-Specialized Multimodal AI Gigapixel pathology VLMs / MLLMs

07 Additional Research Image restoration · on-device efficient vision

PROFILE



Byung Hyun Lee

AI Researcher

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https://hyun1a.github.io

EDUCATION

Ph.D., Electrical & Computer Engineering

Seoul National University | 2021.03 – 2027.02 (expected)

B.S., Electrical Engineering

UNIST | 2015.03 – 2018.08 | GPA 4.04 / 4.30

EXPERIENCE

Visiting Student Researcher, NAVER Cloud

National AI Foundation Model Project | 2025.10 – 2026.02

RESEARCH INTERESTS

Models & Modalities

- Multimodal LLMs
- LLM Reasoning
- Pathology VLMs & MLLMs
- VLA
- Diffusion Models

Methods

- Cross-modal Alignment
- Self-distillation & RL
- Continual Learning
- Concept Erasure

PUBLICATIONS

8
Top AI conference
7 as first / co-first

4
Under review
ICLR 2027 · 3 co-first

4
Workshop / short
CVPR-W · ECCV-W · ICASSP

2
Journal papers
IEEE Access · PLOS One

2
Technical reports
HyperCLOVA X 8B · 32B

1
Patent
Granted in US · KR

PUBLICATIONS FIRST / CO-FIRST AUTHOR

TOP AI CONFERENCE (SELECTED PAPERS)

- Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs **NeurIPS 2026 (Spotlight)***
- Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models **CVPR 2026***
- Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis **ICCV 2025**
- Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation **CVPR 2025***
- Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate **ICLR 2025***
- Doubly Perturbed Task-Free Continual Learning **AAAI 2024 (Oral)**
- Online Continual Learning on Hierarchical Label Expansion **ICCV 2023***

* Equal contribution

UNDER REVIEW · PREPRINT

- HYPERION: Hyperbolic Grounding for Histopathological Slide Representations **ICLR 2027 submitted***
- Self-Distilled Reinforcement Learning for Language-Robust VLA Models **ICLR 2027 submitted***
- S3-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation **ICLR 2027 submitted***
- Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis **Preprint 2026***

JOURNAL

- Towards Accelerating Model Parallelism in Distributed Deep Learning Systems **PLOS One 2023***

SHORT PAPER · DOMESTIC

- Empirically Accelerating Scaled Gradient Projection Using DNN for Inverse Problems **ICASSP 2021**
- Selective Concept Erasing for Safe Diffusion Models (Best Poster Award) **KSPC 2024**

PUBLICATIONS CO-AUTHOR · TECHNICAL REPORTS · PATENTS

CONFERENCE · UNDER REVIEW

- All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters for Specific Degradations **CVPR 2023**
- RePath: Zero-Shot Whole-Slide Tumor Segmentation via Intra-Slide Relational Affinity **ICLR 2027 (sub.)**
- Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation **arXiv 2025**

JOURNAL

- Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation **IEEE Access 2025**

TECHNICAL REPORT · HYPERCLOVA X TEAM

- HyperCLOVA X 8B Omni / 32B Think **2026**

WORKSHOP · DOMESTIC

- Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models **CVPR Workshop 2026**
- Expert Classifier Ensemble-based Post-processing Correction for Unbiased Scene Graph Generation **IPIU 2024**
- Mobile AI & AIM 2022 Challenge Reports (quantized super-resolution; depth estimation) **ECCV Workshop 2022**
- Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction **KSPC 2022**

PATENT

- All-in-one image quality improvement model **US 2026 · KR 2025 (granted)**

RESEARCH OVERVIEW

Multimodal LLMs & Selective Knowledge Updates

Multimodal LLMs
Align & post-train

ALIGNMENT · POST-TRAINING

DCAT
NeurIPS 2026 Spotlight

HyperCLOVA X 8B Omni
NAVER Cloud 2026

S^3-OPSD
ICLR 2027 sub.

MOSAR (VLA)
ICLR 2027 sub.

Expanding
Add knowledge, no forgetting

CONTINUAL LEARNING

HLE
ICCV 2023

DPCL
AAAI 2024 Oral

CoMEL
ICCV 2025

Erasing
Remove target knowledge

CONCEPT ERASURE

CPE
ICLR 2025

GLoCE
CVPR 2025

ETC
CVPR 2026

Domain-Specialized
Gigapixel vision-language

PATHOLOGY VLM · MLLM

HYPERION
ICLR 2027 sub.

Speculative Encoding
Preprint 2026

Tree-of-Experts
MICCAI REG 2025 · 2nd

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Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs

NeurIPS 2026 · Spotlight | Co-first author

MULTIMODAL LLM

A strong vision MLLM lifts weak audio / video MLLMs — closed form, no fine-tuning.

PROBLEM

Weak modalities need **large modality data & compute**; same-modality variants to merge are rare

IDEA

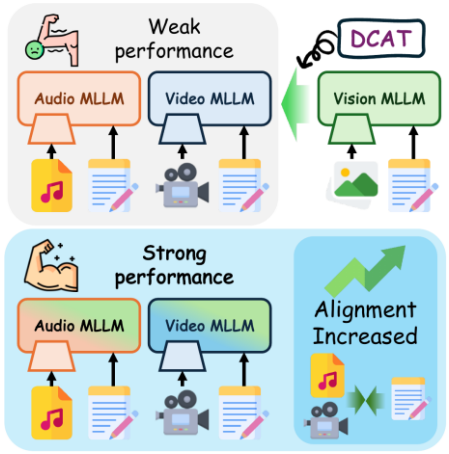
Align = SEO$^{\gamma}$ · SD$^{1-\gamma}$ lower-bounds text–modality MI; raise it in closed form

RESULT

+29.55% audio · **+24.92%** video avg.; beats 6 merging baselines

KEY TECHNIQUES

Donor-anchored cross-modal term · norm-preserving spectral target T_{α} · layer-wise closed form

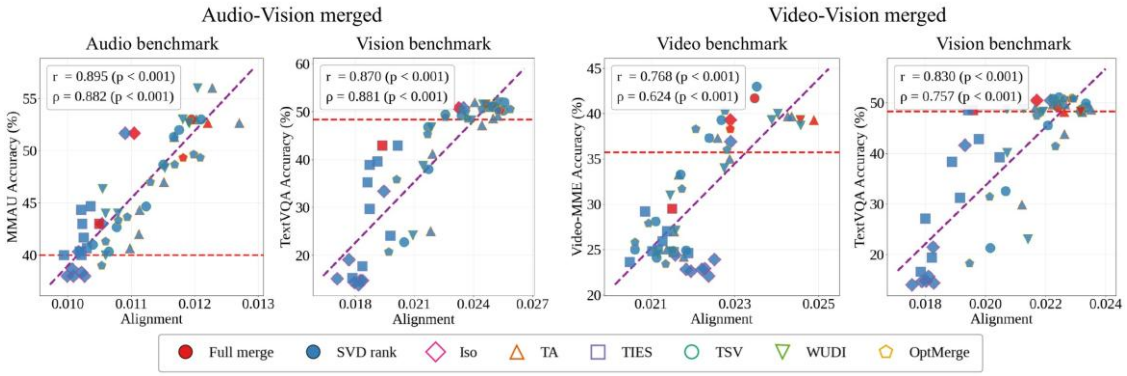


DCAT concept

+29.55% Avg. gain, audio benchmarks

+24.92% Avg. gain, video benchmarks

r = 0.895 Alignment–accuracy correlation



Alignment score vs. performance (merged models)

Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs

NeurIPS 2026 · Spotlight | Co-first author

MULTIMODAL LLM

ALIGNMENT METRIC WITH TOP-k TEXT SUBSPACE(U_k)

$$\mathrm{SEO} = \frac{\|Y_{\mathrm{mod}} U_k^{\top}\|_F^2}{\|Y_{\mathrm{mod}}\|_F^2}$$

Subspace Energy Overlap: share of modality-token energy inside the text subspace

$$\mathrm{SD} = \frac{1}{k}\exp\left(-\sum_{j=1}^{k} p_j \log p_j\right)$$

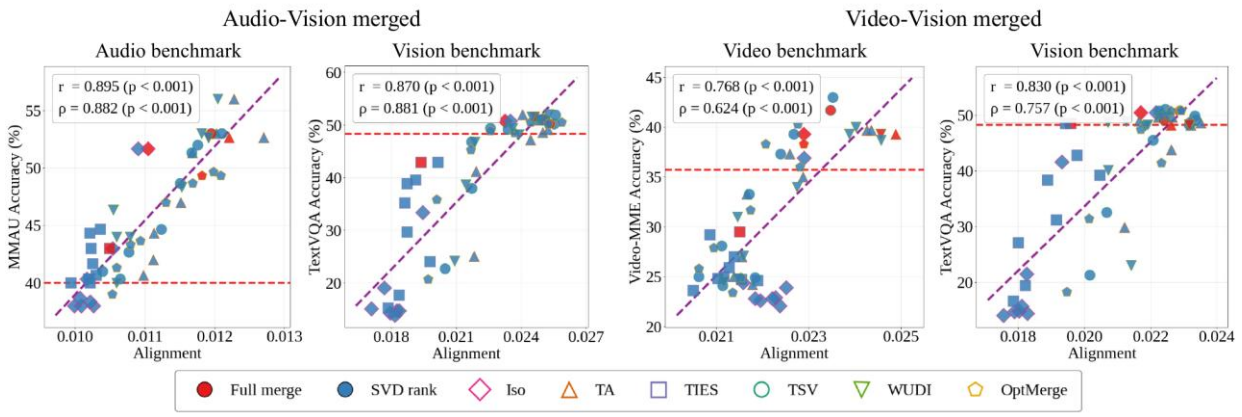
Spectral Diversity: normalized effective rank of $Z = Y_{\mathrm{mod}} U_k^{\top}$ in that subspace

THEOREM · LOWER BOUND ON TEXT–MODALITY MI

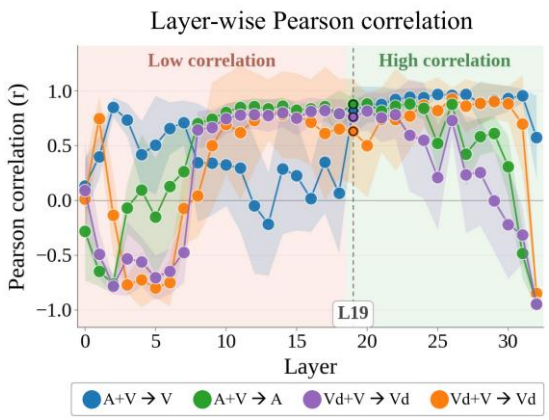
$$I(y;x) \;\geq\; \kappa(\gamma)\log\left(\mathrm{SEO}^{\gamma}\cdot \mathrm{SD}^{1-\gamma}\right) + C_{\delta}$$

Gaussian $z = yU_k^{\top}$, $p_j \geq \delta/k$, $\kappa(\gamma) = k / 2\cdot\min(\gamma, 1-\gamma)$

$$\mathrm{Align}_{\gamma} = \mathrm{SEO}^{\gamma}\cdot \mathrm{SD}^{1-\gamma}$$ → one score; **higher Align ⇒ higher MI bound**



Align vs. accuracy at layer 21 over 114 merged checkpoints



High correlation from L19 on

114
merged checkpoints

r = 0.895
Align vs. MMAU accuracy

$p \ll 10^{-10}$
Align ↑ under merging

Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs

NeurIPS 2026 · Spotlight | Co-first author | Method & results

MULTIMODAL LLM

DCAT · CLOSED-FORM UPDATE

Cross-modal term — recipient & donor anchors on text inputs

$$\mathcal{L}_{\text{cross}}(\tau) = \beta \left\| X_{\text{text}}(\tau - \tau_r) \right\|_F^2 + (1-\beta) \left\| X_{\text{text}}(\tau - \tau_d) \right\|_F^2$$

Self-alignment — norm-preserving target in the text subspace

$$\mathcal{L}_{\text{self}}(\tau) = \left\| X_{\text{mod}}(\theta_0 + \tau) - T_\alpha \right\|_F^2, \quad T_\alpha P_k^\perp = 0, \quad \|T_\alpha\|_F = \|Y_{\text{mod}}\|_F$$

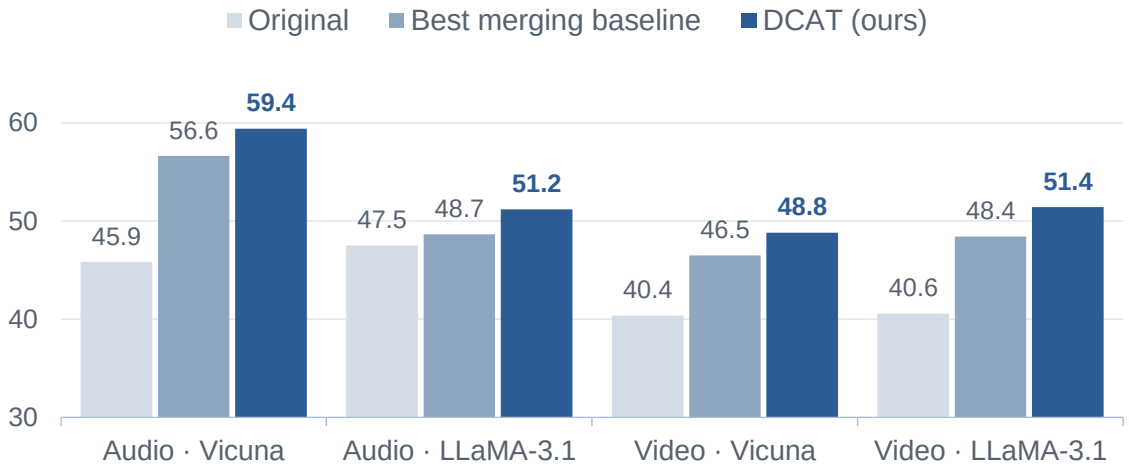
Objective — least squares ⇒ **closed form** per linear projection

$$\tau^\star = \arg\min_\tau \; \mathcal{L}_{\text{self}}(\tau) + \eta \, \mathcal{L}_{\text{cross}}(\tau)$$

Adaptive spectrum — set per projection from recipient SEO / SD

$$\alpha = \text{Sigmoid}\left(\frac{\log \text{SEO}_r}{\log \text{SD}_r + \epsilon} \right)$$

AVERAGE ACCURACY (%) on 9 BENCHMARKS



Best in all 4 settings vs. 6 merging baselines

INDUSTRY EXPERIENCE

NAVER Cloud · HyperCLOVA X 8B Omni / 32B Think

Visiting Student Researcher

NAVER Cloud | Oct 2025 – Feb 2026

HyperCLOVA X 8B Omni & HyperCLOVA X 32B Think

Post-training

CONTRIBUTION

- ✓ Training & inference **stability**
- ✓ **+20% throughput** (dynamic batching)

Inference & deployment

CONTRIBUTION

- ✓ Stable at **~2× faster inference**
- ✓ Code for **online deployment**

~2×
faster inference, kept stable

+20%
training tokens per unit time

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04 Expanding: Continual Learning — Adding knowledge without forgetting

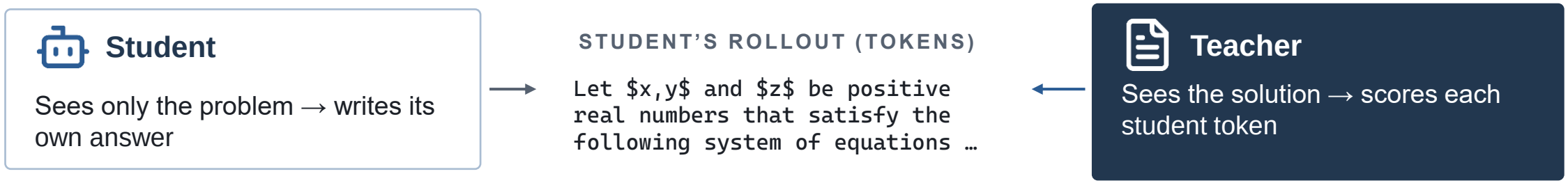
05 Erasing: Concept Erasure — Unlearning targets, keeping the rest

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07 Additional Research — Image restoration · on-device efficient vision

ON-POLICY SELF-DISTILLATION (OPSD)

No external teacher: the **same LLM, shown the solution**, grades its own attempt token by token.



Challenges of GRPO/OPSD & **Our solution**

✕ GRPO (RL)

Needs **many rollouts** → slow, expensive

✕ OPSD

Dense supervision; **can be unstable**

✓ S^3-OPSD (ours)

Key tokens only → **13× faster than GRPO**, AIME24 **+10.0**

Why it matters Teacher-free, **dense token-level credit assignment** — a cheap complement to RLVR for LLM / MLLM post-training.

S^3-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation

ICLR 2027 (submitted) | Co-first author

POST-TRAINING

Better LLM reasoning without an external teacher — stable and fast.

PROBLEM

Self-distillation is **costly & unstable**; RL needs **many rollouts**

IDEA

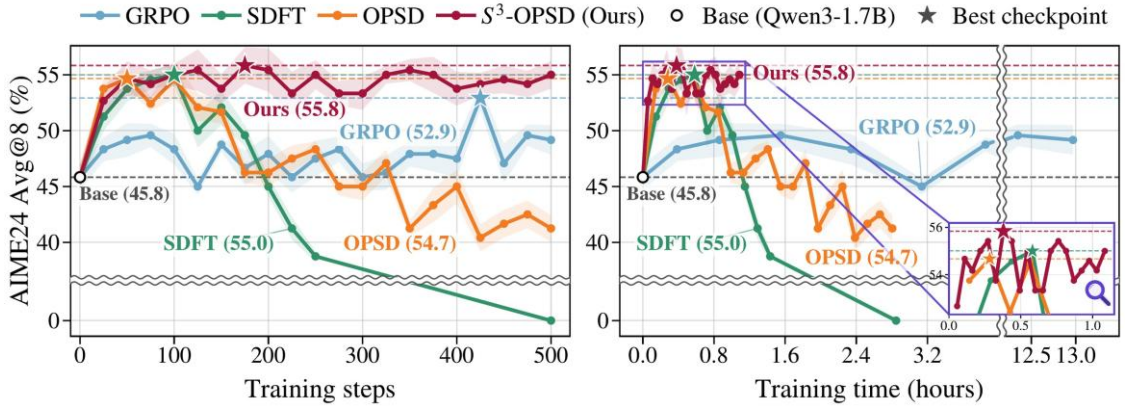
Supervise only the **important tokens**, with **small, stable updates**

RESULT

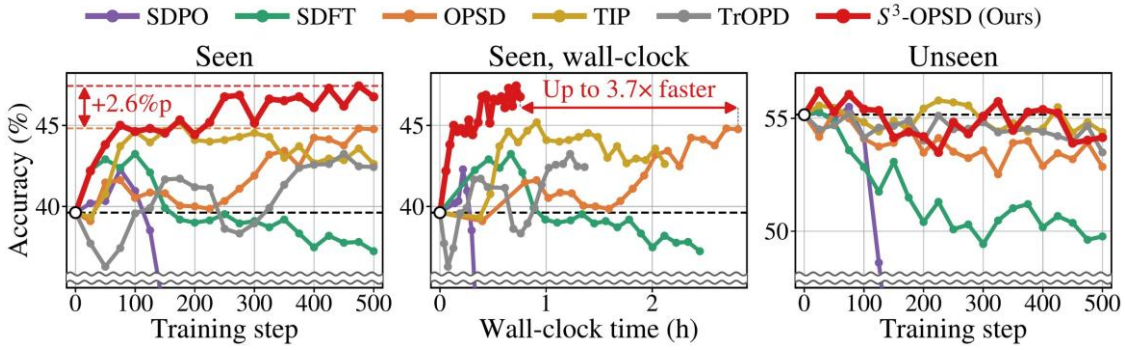
AIME24 **55.8 (+10.0)** · up to **13× faster** than RL (GRPO)

KEY TECHNIQUES

Advantage-variance state selection · SGD-induced sparse updates · smoothed targets



AIME24 vs. training steps (left) and wall-clock time (right)



Multi-task learning over seven domains

Self-Distilled Reinforcement Learning for Language-Robust VLA Models

ICLR 2027 (submitted) | Co-first author

POST-TRAINING

A robot that succeeds with one instruction can fail with a paraphrase of it.

PROBLEM

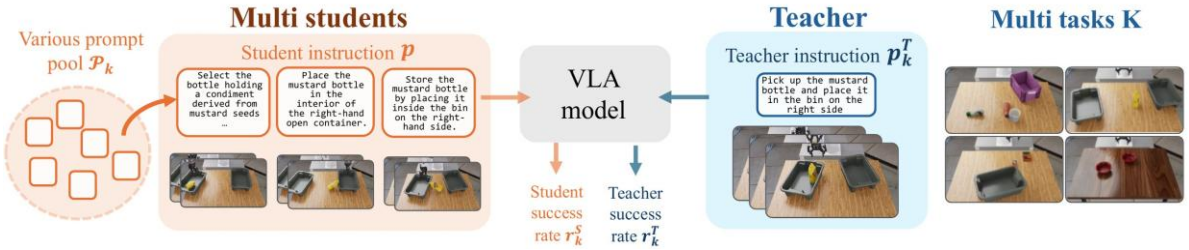
Paraphrased instructions → **very different success rates**

IDEA

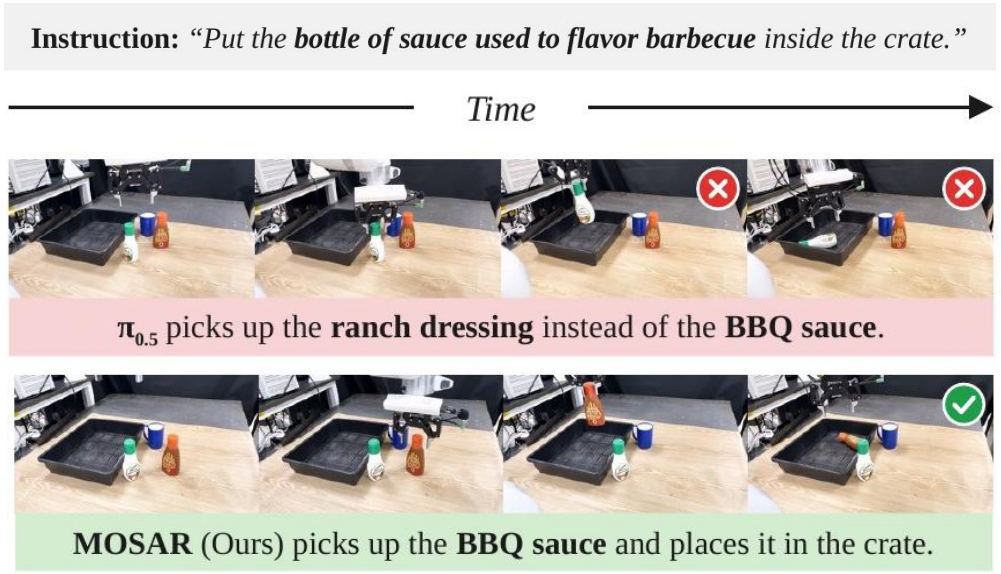
OPSD from the **best-phrased prompt** to its paraphrases; **RL** keeps improving the best

RESULT

Robust to paraphrases; unseen prompts & tasks preserved · verified on a **real Franka robot**



Overview of MOSAR



Real robot: only MOSAR picks the correct object

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WHY KNOWLEDGE UPDATES

Foundation models now power products on three fronts — and each must keep updating its knowledge after deployment.

Agentic / Physical AI

Assistants · agents · robots

Online knowledge addition

Adapt quickly to new tasks, tools, environments and instructions

Vertical AI

Domain-specialized models

Domain knowledge addition

New domains, modalities, languages and tasks

Visual Generative AI

Text-to-image / video generation

Harmful knowledge deletion

Remove copyrighted, private or unsafe concepts on demand

DEPLOYMENT

Deployed models must adapt — without full retraining

Online knowledge addition agents & robots

Domain knowledge addition new modalities & domains

RESEARCH

- **Catastrophic forgetting** in foundation models
- **Stability–plasticity** trade-off
- **Precise, scalable unlearning**

THE CHALLENGE: Update knowledge without forgetting

1 Full retraining

Retrain on all the data



Highly inefficient and costly

2 Naive knowledge update

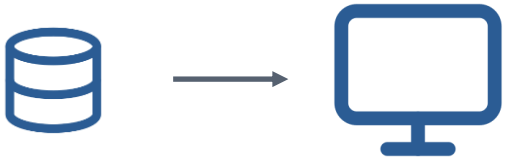
Fine-tune only on the new data



Forgets existing knowledge (stability–plasticity dilemma)

3 Selective knowledge update

Update only the target, keep the rest



Efficient and minimal forgetting

Example Adding a new skill, language or modality to a deployed assistant: no retraining from scratch (1), no forgetting (2) — update only what is new (3).

Online Continual Learning (CL) on Hierarchical Label Expansion

ICCV 2023 | Co-first author

EXPANDING

Learns finer classes as labels get refined over time ("bird" → "sparrow").

PROBLEM

Labels grow coarse → fine, but standard continual learning **ignores the hierarchy**

IDEA

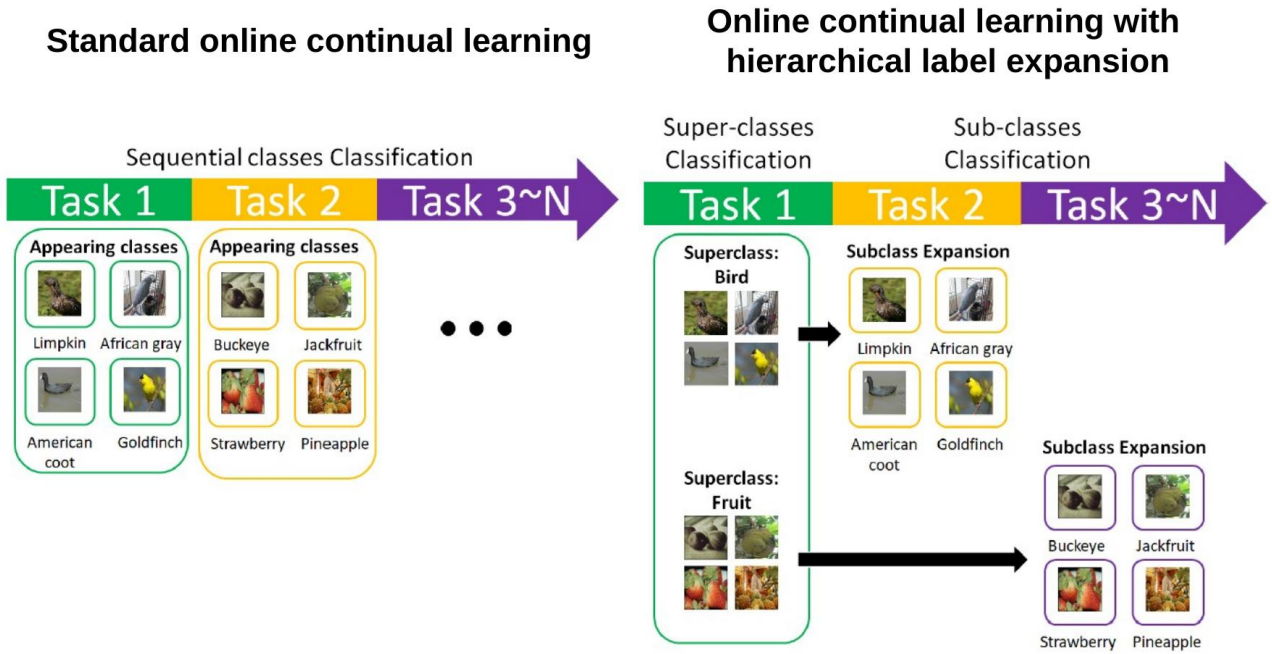
Use the **hierarchical memory**: pseudo-labels per branch + memory favoring new classes

RESULT

Best any-time accuracy on 3 benchmarks

KEY TECHNIQUES

Hierarchy-aware pseudo-labeling · branch-aware memory · flexible memory sampling

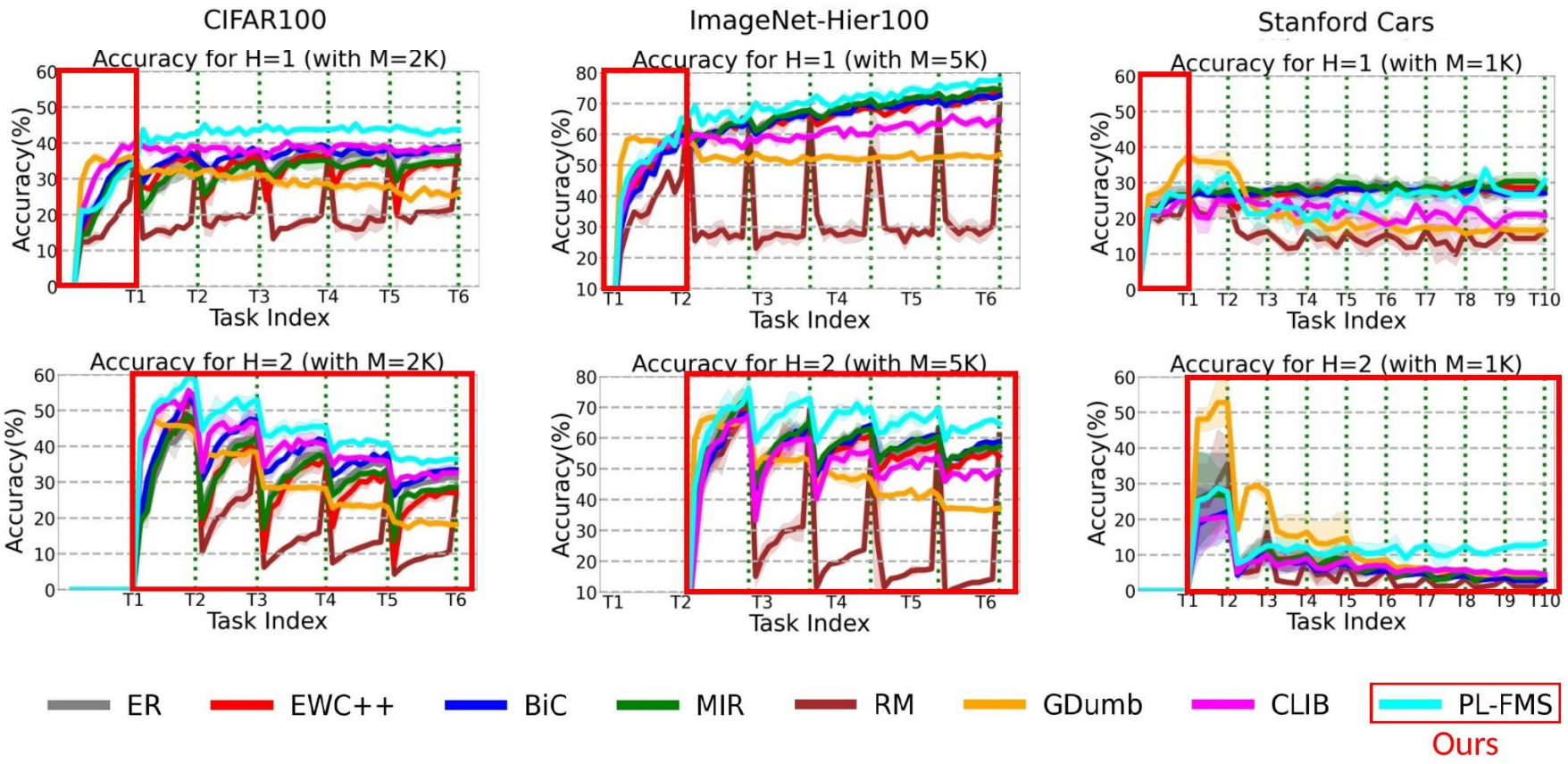


Standard online CL (left) vs. hierarchical label expansion (right)

Online Continual Learning on Hierarchical Label Expansion

ICCV 2023 | Co-first author

EXPANDING



Any-time accuracy (top: super-class, bottom: sub-class)

Doubly Perturbed Task-Free Continual Learning

AAAI 2024 · Oral | First author

EXPANDING

Modeling the uncertainty of unseen future data → less forgetting, faster adaptation.

PROBLEM

Streaming data, no task boundaries: **today's update can hurt future learning**

IDEA

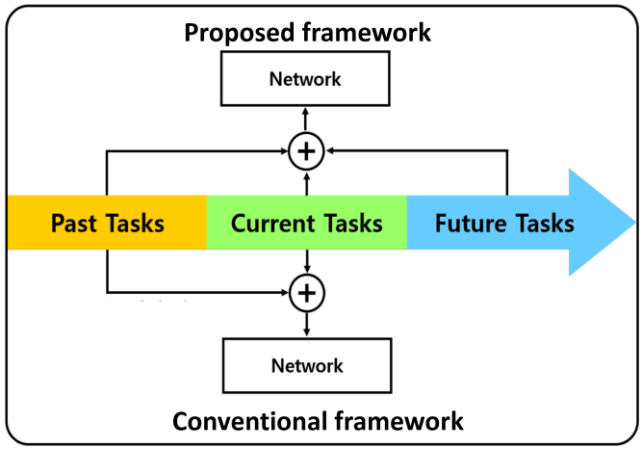
Train with random **perturbations of inputs and weights** (flatter minima)

RESULT

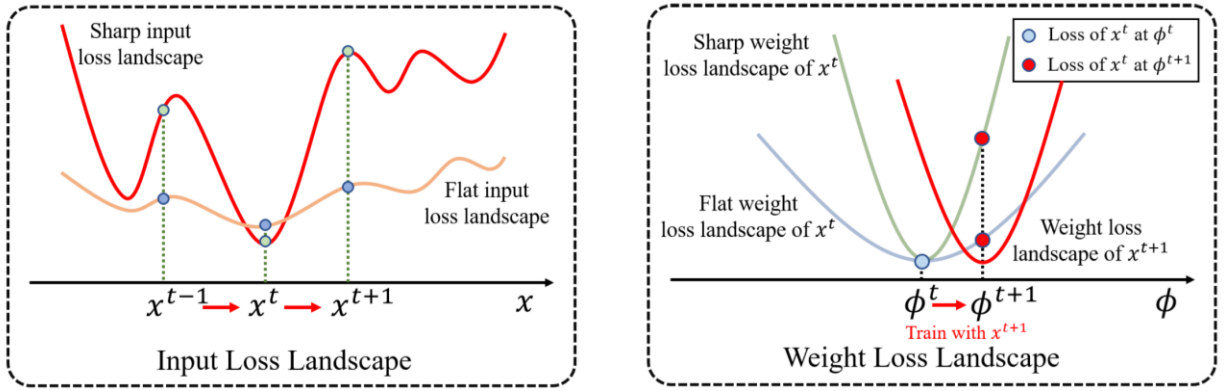
Best any-time accuracy on 3 benchmarks · **AAAI Oral**

KEY TECHNIQUES

Perturbed function interpolation · stochastic classifiers · perturbation-aware memory



Conventional vs. ours: also prepare for future tasks

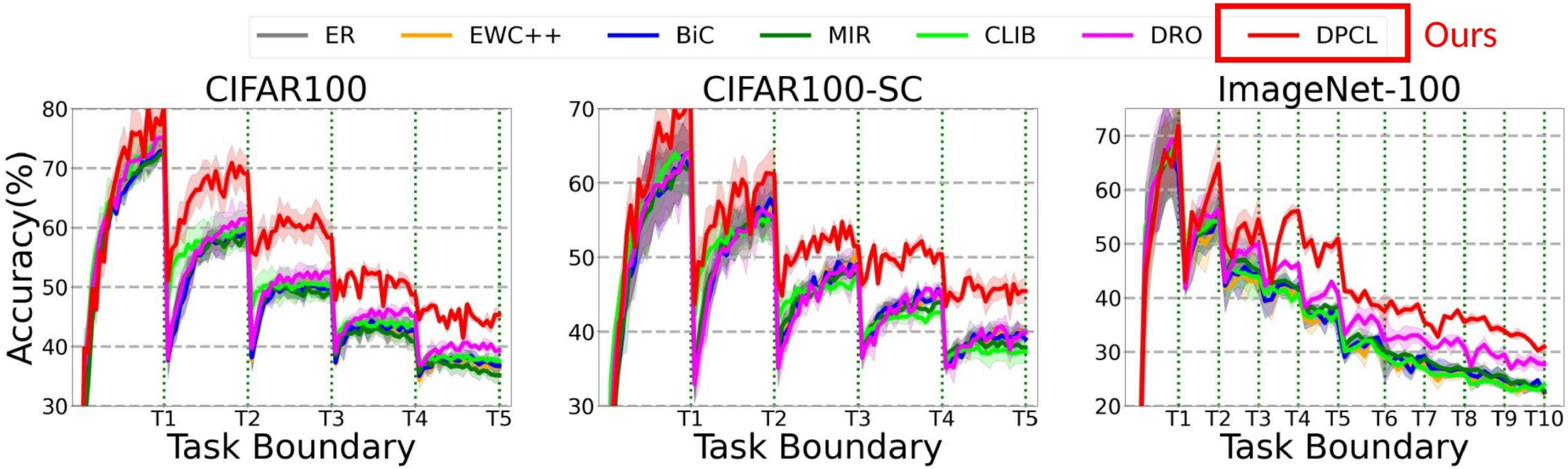


Sharp vs. flat loss landscapes (input / weight)

Doubly Perturbed Task-Free Continual Learning

AAAI 2024 · Oral | First author

EXPANDING



Any-time accuracy in class-incremental learning

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CONCEPT ERASURE

REMOVE ONE THING, KEEP EVERYTHING ELSE

Generative models can reproduce **copyrighted, private or unsafe** content.

Goal: delete **only the target** — no full retraining, no damage to the rest.

STEP 1 · PRECISE

CPE

ICLR 2025 · Co-first author

A **gate** fires only for target concepts

✓ **50 celebrities / 100 styles** erased

STEP 2 · LOCALIZED

GLoCE

CVPR 2025 · Co-first author

Erase only the **target's region** — no training

✓ **Other concepts** in the image intact

STEP 3 · SCALABLE

ETC

CVPR 2026 · Co-first · Champion Award

One **mixture-of-experts** module for all targets

✓ **2,072 concepts** at once

NAIVE vs. SELECTIVE UPDATE

✗ **Naive fine-tuning:** erases the target but **damages related concepts** and image quality

✓ **Selective update:** target gone, **everything else preserved** — measured on both sides

Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate

ICLR 2025 | Co-first author

ERASING

Erases target concepts while keeping the others intact.

✕ **PROBLEM**

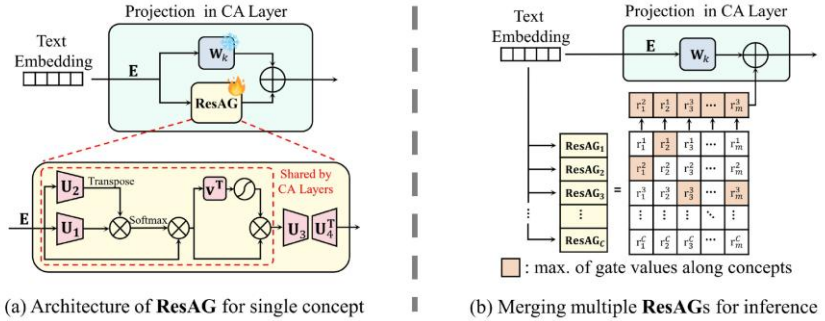
Linear fine-tuning **cannot erase and preserve at the same time**

IDEA

Add a **residual attention gate** that fires only for the targets

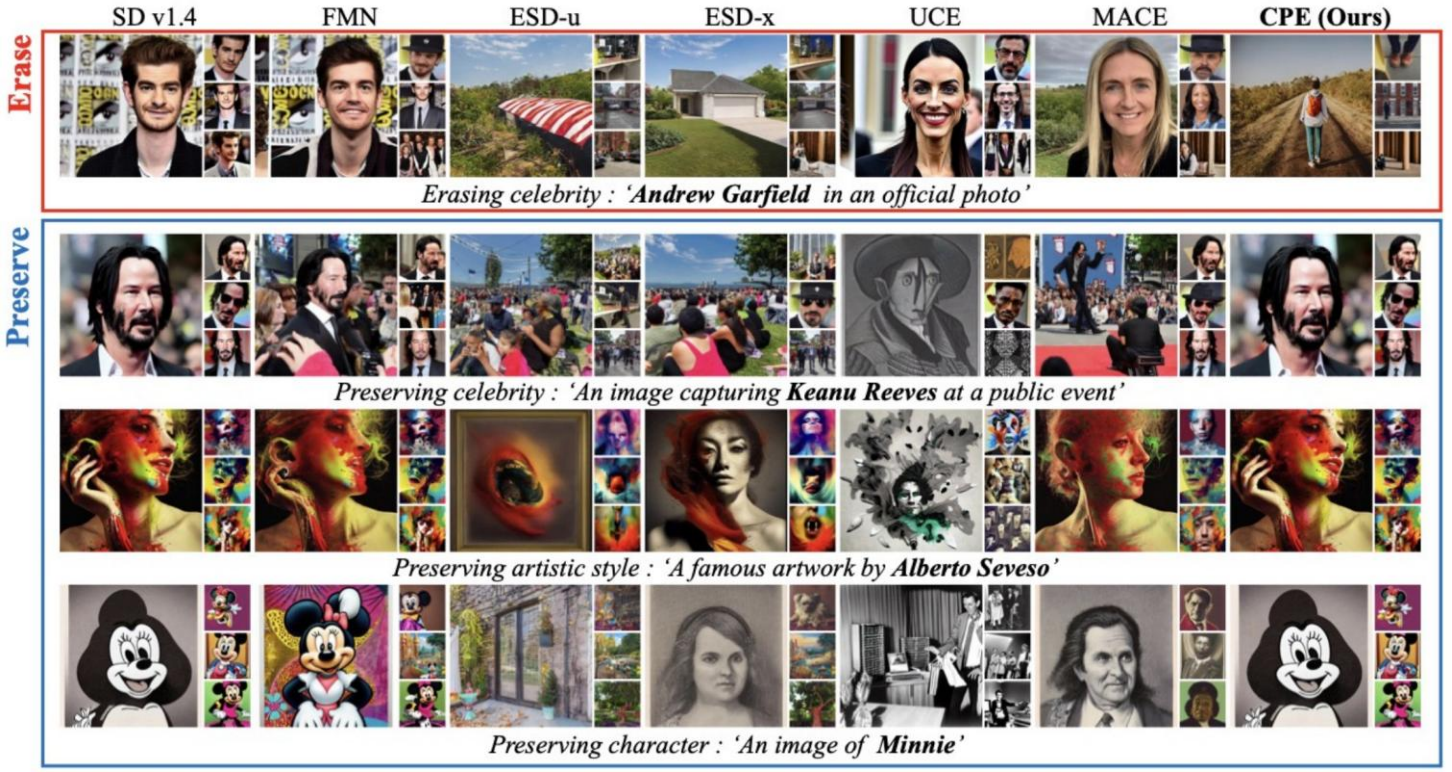
✓ **RESULT**

Erase **50 celebrities / 100 styles**, others near-original



(a) Architecture of **ResAG** for single concept

(b) Merging multiple **ResAG**s for inference



Erasing 50 celebrities: erased vs. preserved

Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation

CVPR 2025 | Co-first author

ERASING

Erases only the target's region of an image — training-free.

✕ **PROBLEM**

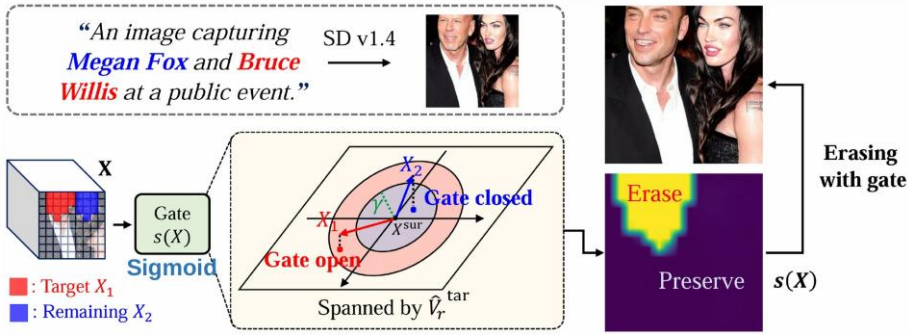
Prior methods **damage the whole image** to remove one concept

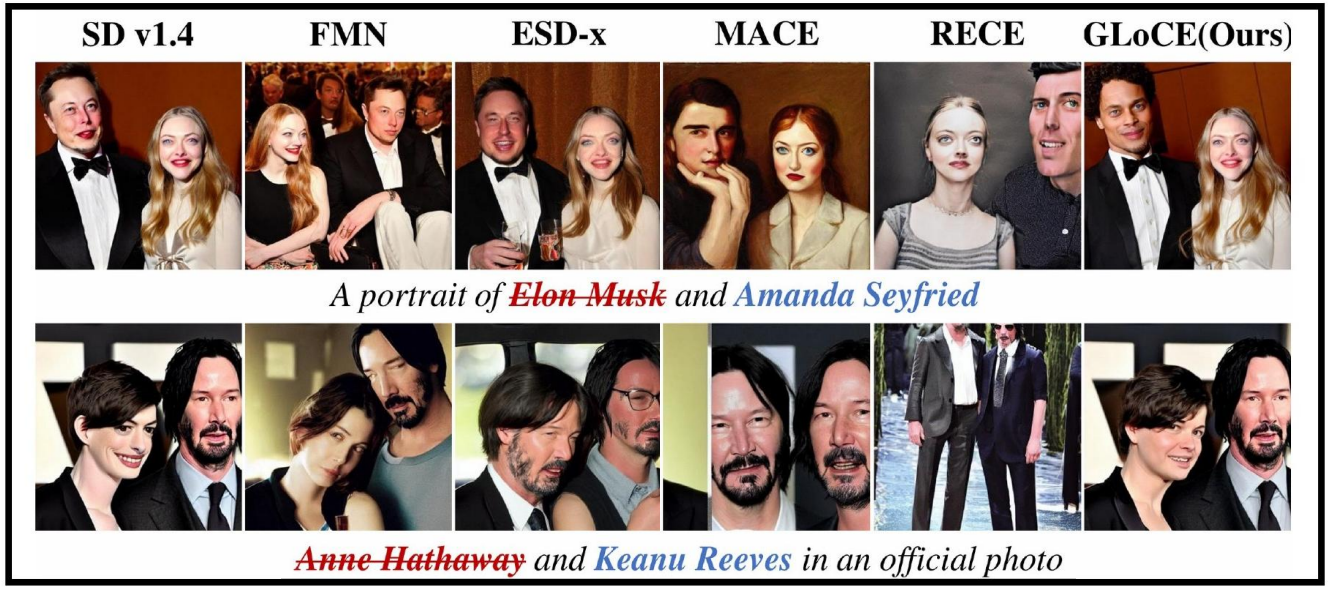
IDEA

A gate **opens only on the target region**; the rest passes untouched

✓ **RESULT**

Localized erasure on **SD v1.4 / SD v3**, **no training**





Localized celebrity erasure

Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models

CVPR 2026 | Co-first author · Compute Transparency Champion · Popular Science Korea

ERASING

Erases 2,072 concepts at once while keeping generation quality.

PROBLEM

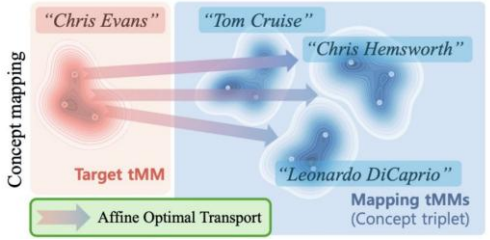
Prior methods **break down beyond a few hundred** concepts

IDEA

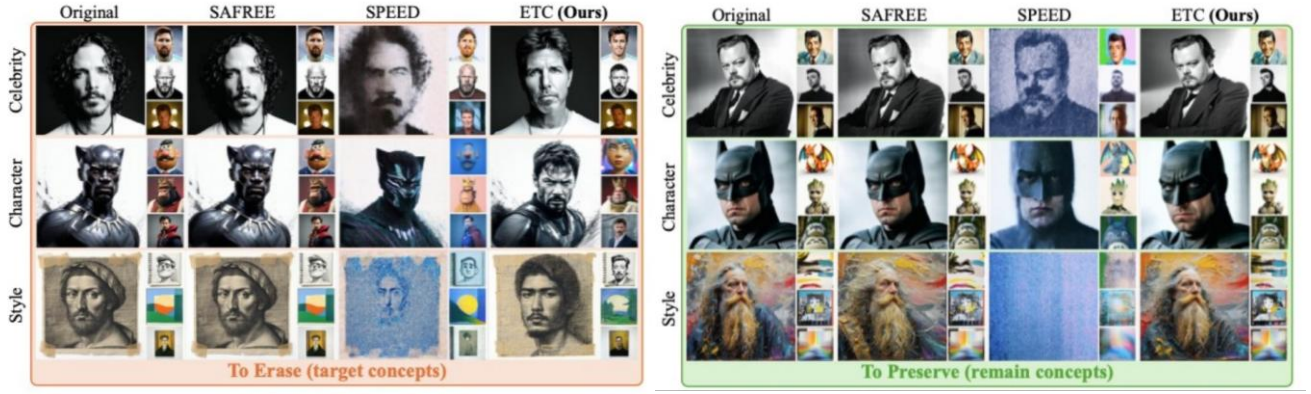
Map each target to a **safe nearby concept**; **one module** handles all

RESULT

2,072 concepts (SD v1.4) · **515** (SD v3.5)



Concept mapping via affine optimal transport



Targets erased (left) · other concepts preserved (right)

MORE ON GENERATIVE MODELS

Unlearning · Semantic Binding · Safe Diffusion

Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models

CVPR Workshop 2026 · Co-author

- Removes outputs **prompts cannot specify** (e.g., a person's face)
- **Image-editing surrogates** + timestep-aware weighting
- Validated on **SD3** and **DDPM-CelebA**

Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation

arXiv 2025 · Co-author

- Attributes often **bind to the wrong object**
- Analyzes the **geometry of text-token embeddings**
- Uses it for **stronger semantic binding**

Selective Concept Erasing for Safe Diffusion Models

KSPC 2024 · First author · Best Poster

- Erases **unsafe concepts**, keeps the rest
- **Best Poster Award**
- Early work behind **CPE, GLoCE and ETC**

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04 Expanding: Continual Learning — Adding knowledge without forgetting

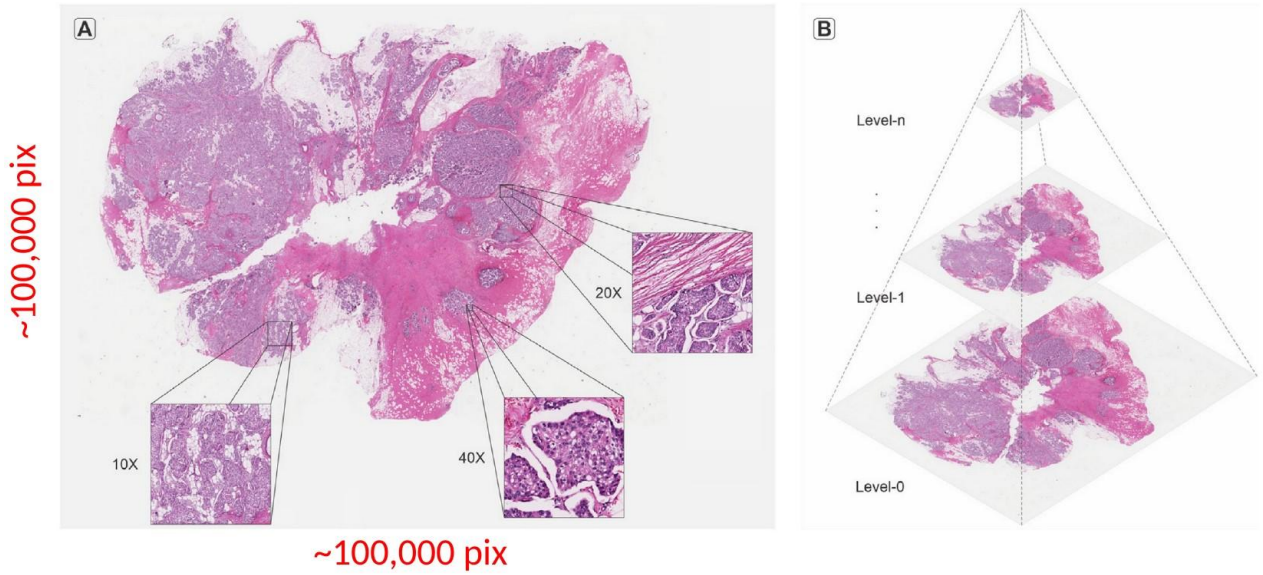
05 Erasing: Concept Erasure — Unlearning targets, keeping the rest

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07 Additional Research — Image restoration · on-device efficient vision

WHOLE SLIDE IMAGES WHAT THE DATA LOOKS LIKE

A tissue slide scanned at cell-level resolution — **one image ≈ 100,000 × 100,000 pixels**.



A whole slide image (left) and its zoom pyramid (right)

~10 B pixels in one slide (several GB)

10K–100K patches the slide is cut into

GIGAPIXEL PATHOLOGY

BOTTLENECKS & MY CONTRIBUTIONS

STEP	BOTTLENECK	PROPOSED METHOD	RESULT
Encode patches	**~99% of inference time**	**Speculative Encoding** draft cheaply, verify selectively	**3× faster**, 6× less memory
Aggregate & Locate	**Only slide labels** **No pixel labels**	**CoMEL** new organs, no stored data	**+23.4%** localization IoU
		HYPERION no pixel labels needed	Dice **0.143 → 0.463**
Report	**Strict format**, < 10K slides	**Tree-of-Experts** one small expert per branch	**HYPERION+MedGemma**

Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis

ICCV 2025 | First author

DOMAIN-SPECIALIZED AI & EXPANDING

Learns new organs one by one without storing patient data — and keeps tumor localization.

PROBLEM

Storing past slides is **costly & risky**; prior methods **silently forget localization**

IDEA

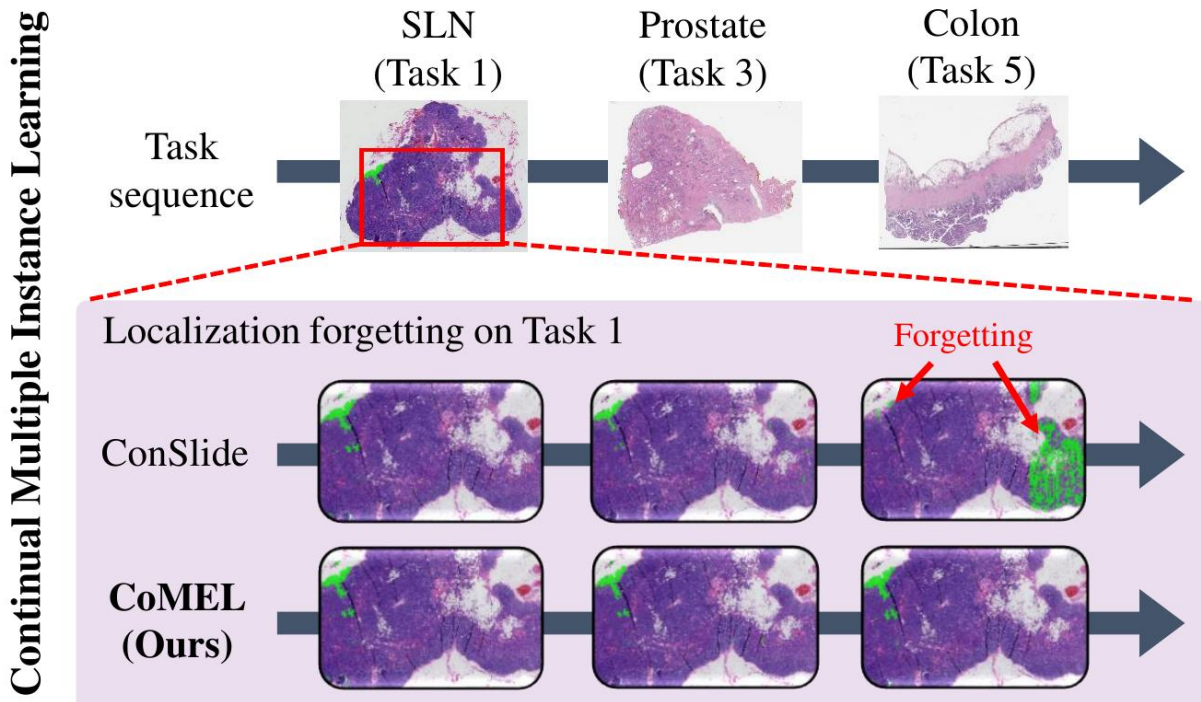
Add new knowledge in directions that **do not overwrite the old** (orthogonal LoRA)

RESULT

Up to **+11.0% slide accuracy** and **+23.4% localization IoU**

KEY TECHNIQUES

Grouped double-attention MIL · bag-prototype pseudo-labels · orthogonal weighted LoRA

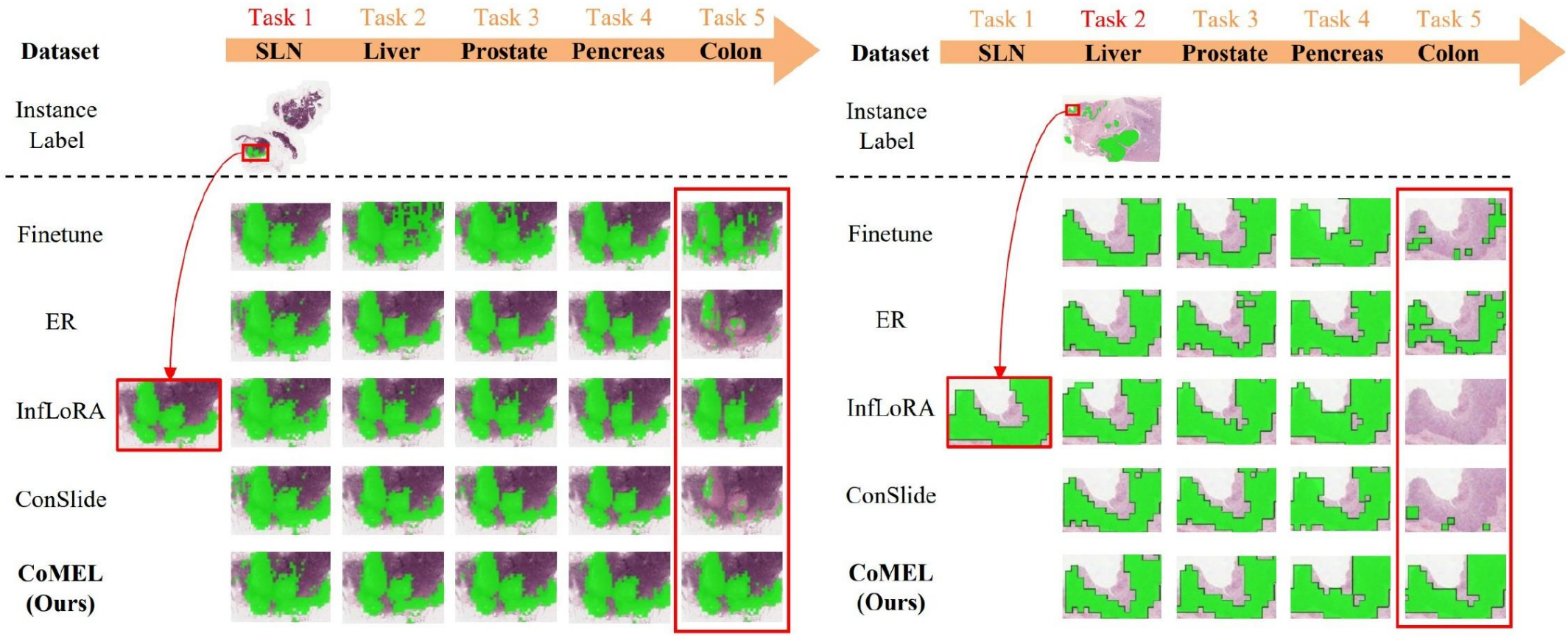


ConSlide forgets Task-1 tumors; CoMEL keeps them

Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis

ICCV 2025 | First author

DOMAIN-SPECIALIZED AI & EXPANDING



Tumor localization over the task sequence

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI

Training a model to point at the evidence for each pathology concept.

PROBLEM

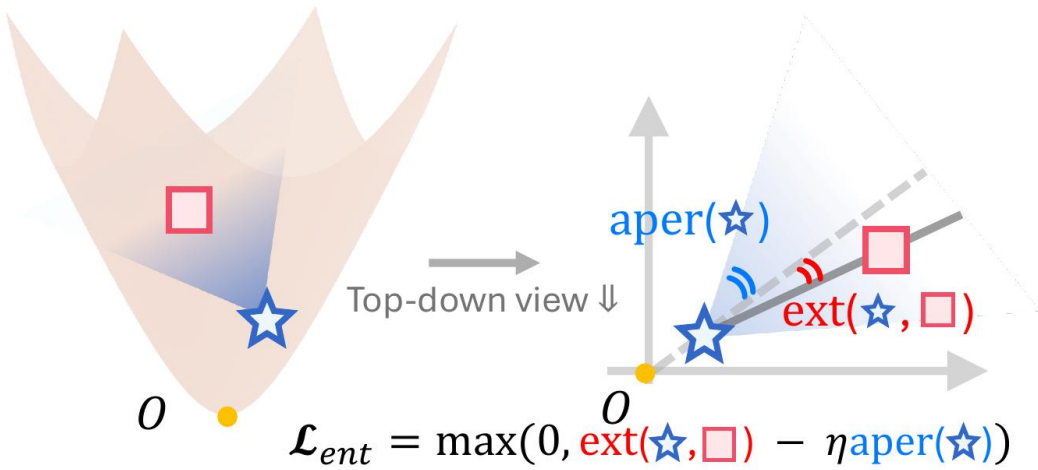
Slide encoders match slide ↔ report but **cannot show which region supports each concept**

IDEA

Fine-tune the slide encoder so slide & text embeddings form a **tree** (hyperbolic space)

RESULT

Zero-shot tumor segmentation: avg. Dice **0.143 → 0.463**



Hyperbolic space ≈ a tree: general concepts at the center, specific evidence at the edge

KEY TECHNIQUES

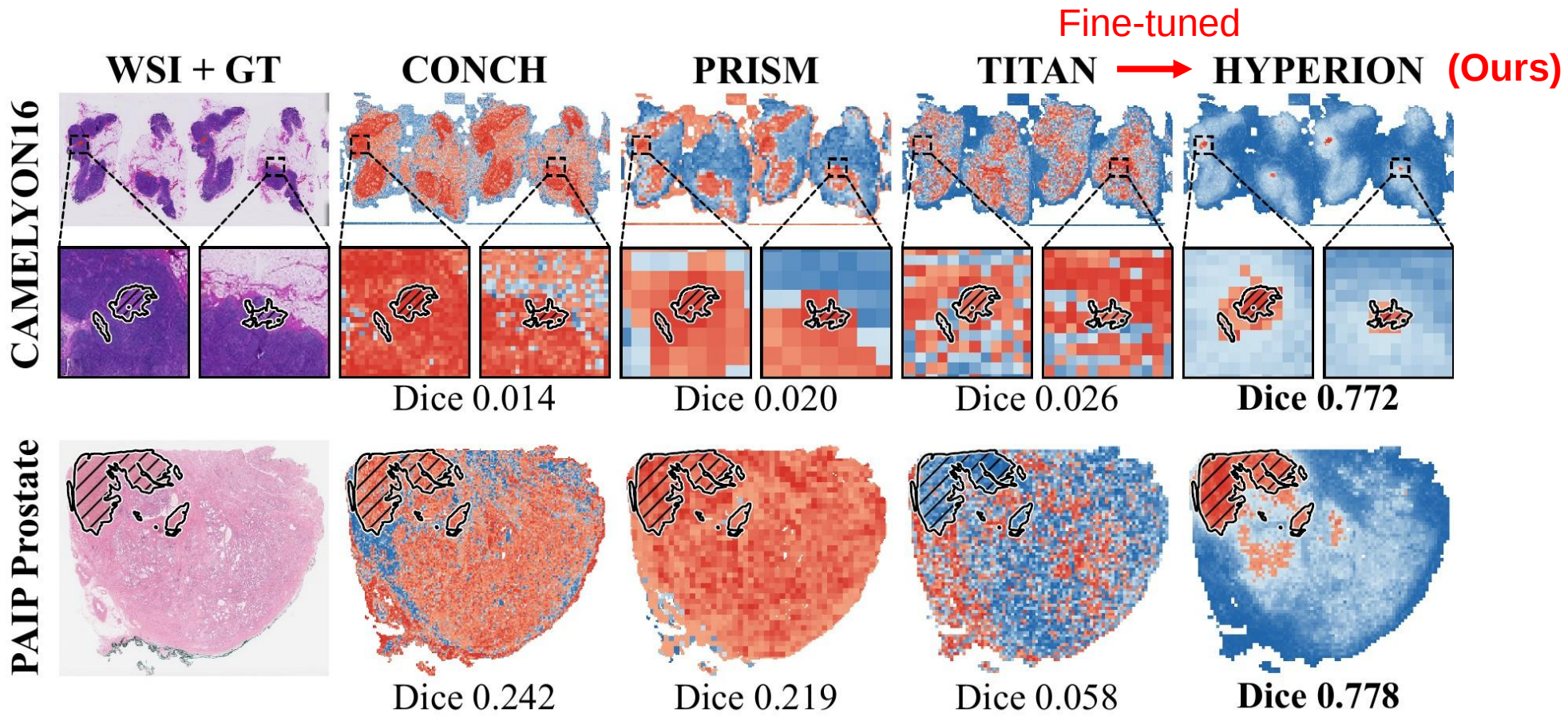
Lorentz-space contrastive + entailment loss · text-conditioned Einstein midpoints · norm propagation

- **Flat (Euclidean)** embeddings squeeze the tree → **weak grounding**
- **HYPERION** keeps the tree → each sentence finds its region

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI

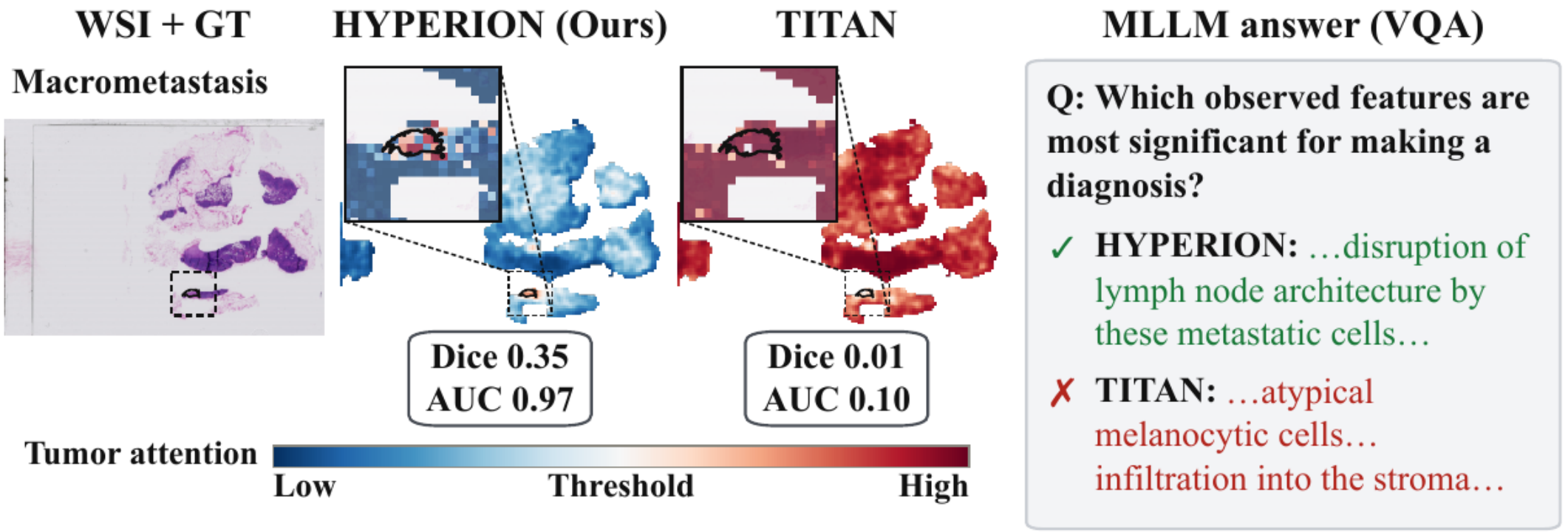


Zero-shot tumor segmentation vs. pathology foundation models

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI



MedGemma-4B WSI MLLM: HYPERION grounds the metastasis and answers correctly; TITAN fails

Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis

Preprint 2026 | Co-first author

DOMAIN-SPECIALIZED AI

Big-model patch encoding is the bottleneck — let a small model decide where to look.

PROBLEM

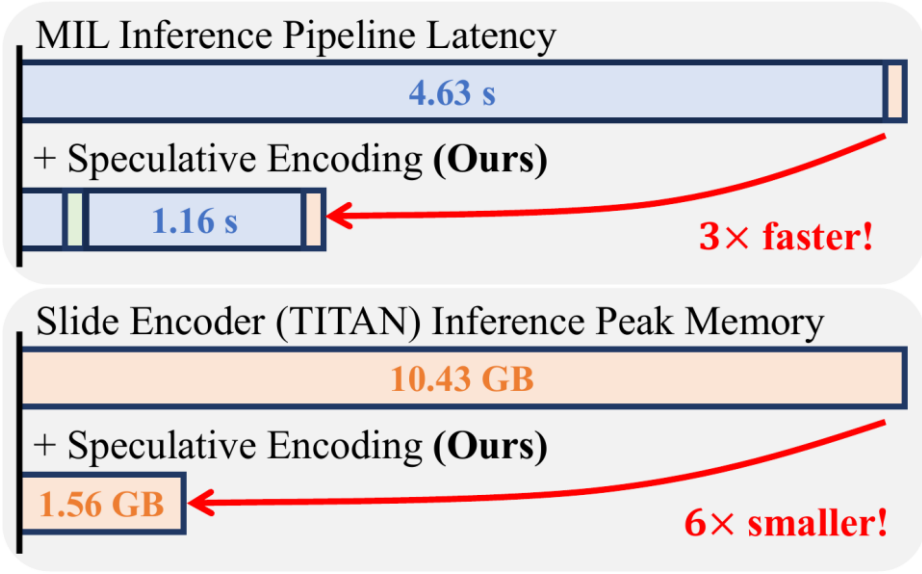
Patch encoding takes **~99% of inference time**; aggregation sets **peak memory**

IDEA

Draft all patches with a **small encoder**, re-encode only the **informative** ones

RESULT

3× faster · 6× less memory · >99% accuracy kept



Latency & peak memory: baseline vs. ours

KEY TECHNIQUES

Relational distillation · diversity + discriminativity patch selection · **1.80× faster WSI-LLaVA MLLM**

3×	6×	>99%
faster inference	**less peak memory**	**accuracy kept**
4.63 s → 1.16 s	10.4 → 1.6 GB	across 12 models

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IMAGE RESTORATION & EFFICIENT VISION

Low-level vision · On-device AI

CVPR 2023 · Co-author

All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters

- One model restores **unknown degradation types**
- Granted **US (2026) and KR (2025) patents**

→ *Robustness to unknown corruptions*

ECCV Workshop 2022 · Challenge participant

Mobile AI & AIM 2022 Challenges — Quantized Super-Resolution on Mobile NPUs · Depth Estimation on Mobile

- Quantized **super-resolution on mobile NPUs**
- Co-author of the **challenge reports**

→ *On-device inference on mobile NPUs*

ICASSP 2021 · First author

Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems

- Speeds up **inverse-problem solvers** with a network

→ *Learned acceleration of iterative solvers*

KSPC 2022 · Co-author

Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction

- **Uncertainty-aware** low-dose CT reconstruction

→ *Uncertainty-aware reconstruction*

CONCLUSION

01 Cross-modal alignment for multimodal & omnimodal LLMs

DCAT (NeurIPS 2026 Spotlight): MI-grounded alignment score and **closed-form transfer** from vision to audio / video MLLMs; post-training of **HyperCLOVA X 8B Omni** at NAVER Cloud

02 Efficient post-training & selective knowledge updates

S^3-OPSD and **MOSAR** for LLM reasoning and VLA robustness; **continual learning** (ICCV · AAAI Oral · ICCV) and **concept erasure** up to 2,072 concepts (ICLR · CVPR · CVPR)

03 From theory to large-scale validation & leadership

Gigapixel pathology VLMs / MLLMs (HYPERION, Speculative Encoding); **project leader** for SD Biosensor, MSIT · NIPA and KETI; **team leader** for MICCAI 2025 REG (2nd)

Next Building omnimodal LLMs that **align new modalities efficiently** and **keep learning without forgetting**.

Thank you

Byung Hyun Lee · AI Researcher

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